Deepfake Detection Using Deep Learning Model

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Project Aims

- **The Problem:** Evolving AI tools allow for effortless manipulation, creating highly realistic deepfakes that undermine the credibility of digital media in law, politics, and finance.

- **Aim:** To research and develop a demo software system capable of accurately distinguishing between real and deepfake audio-visual media using deep learning.

- **Target Scope:** Focusing on facial reenactment (where speech/lip movements are altered) rather than simple face-swaps.

Project Objectives

- Explore deepfake generation technologies and creation methods.
- Analyse the real-world impacts across sectors and evaluate current forensic solutions.
- Conduct quantitative research using secondary analysis on the AV-Deepfake1M++ dataset.
- Design and implement a demo software that detects and differentiates deepfake media using deep learning.
- Evaluate model accuracy, precision, and recall regarding audio-visual manipulation.
- Reflect on outcomes, challenges, and suggest future software developments.

Why is this a Computing problem?

- Detecting synthetic media requires modelling high-dimensional data through multi-layered neural networks (Deep Learning) to identify patterns simpler algorithms miss.

- The AV-Deepfake1M++ dataset is approximately 1.4 TB. Efficiently processing this requires advanced subset strategies and computational management.

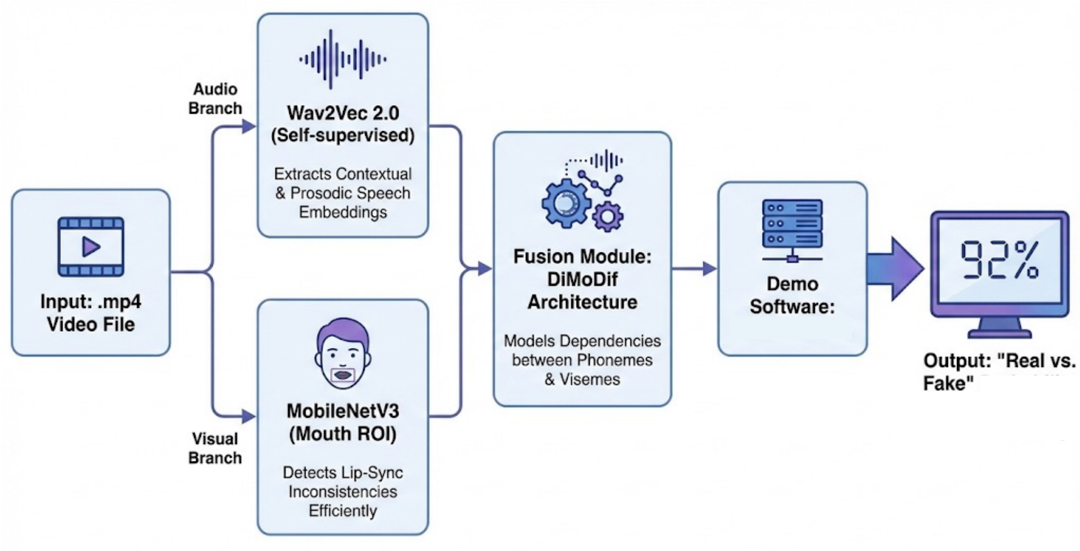
- Generative models (like NeRF and Diffusion) currently outpace detection systems; solving this requires innovative spatio-temporal architectures.

Literature Review Findings

- Evolution: Deepfakes have moved through three phases:
  - Visual Fidelity: Simple face-swaps using Autoencoders/GANs.
  - In-the-Wild: Multi-subject manipulations in uncontrolled environments.
  - Multimodal: Neural generation affecting both audio and video (e.g., lip-sync errors).
- The Gap: Most existing models are vision-centric or rely on simple feature concatenation, failing to capture the fine-grained temporal relationship between speech and lip motion.
- Key Finding: Detectors often overfit to specific datasets and fail against unseen generators, necessitating a focus on generalisation.

What is your proposed solution?

- A multimodal deep learning pipeline prioritising cross-modal coherence.



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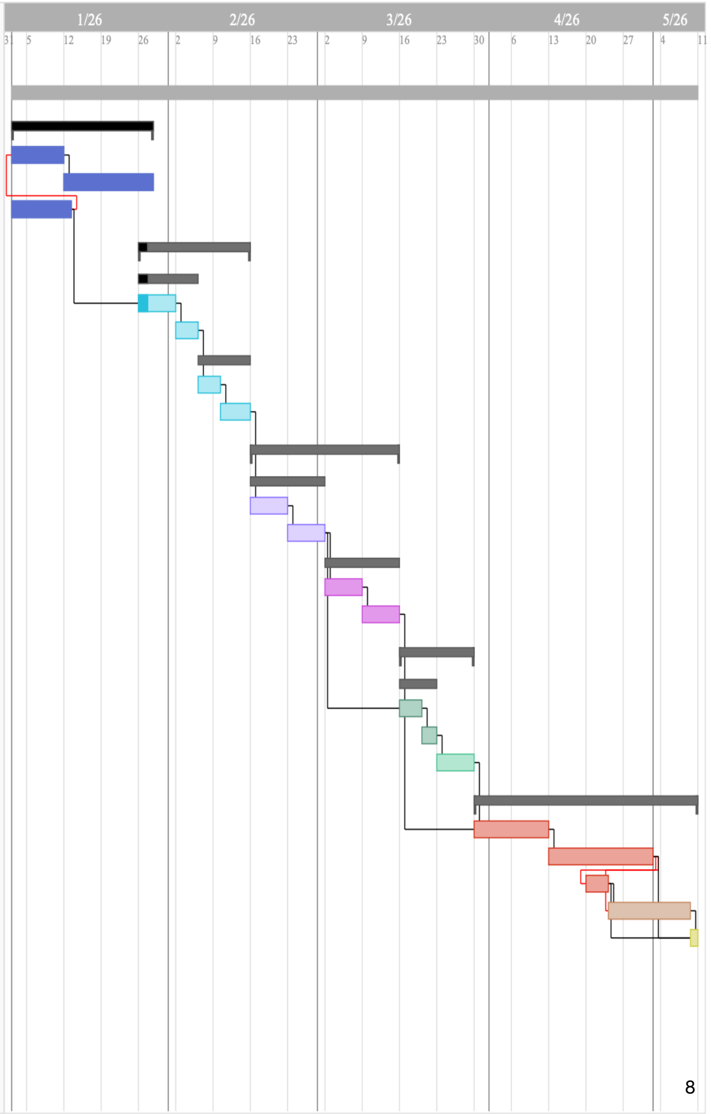
How will you evaluate your solution?

- Methodology: Quantitative analysis using Binary Cross-Entropy loss.
- Metrics: Primary metric is Accuracy, supported by Precision, Recall, and F1-score to assess class balance.
- Validation Strategy: Due to the 1.4 TB dataset size, a subset-based strategy (training/validation splits) is used for hyper-parameter tuning and early stopping.
- Limitations: Evaluation focuses on clip-level classification; frame-level temporal localisation is deferred due to hardware constraints.

Project Planning and Methodology

- **Methodology:** Quantitative research via secondary data analysis using a kanban based incremental approach.

Deepfake Detection Using Deep Lea...	start	end	25%
Planning and Design	**01/01/26**	**01/27/26**	**100%**
Finalise specification and models	01/01	01/09	100%
Viva preperation	01/12	01/27	100%
Acquire dataset	01/01	01/12	100%
Implementation Phase I	**01/26/26**	**02/13/26**	**8%**
Dataset Preparation	**01/26/26**	**02/04/26**	**16%**
Subset Extraction (AV-Deepfake1M++))	01/26	01/30	25%
Data preprocessing	02/02	02/04	0%
Feature Extraction	**02/05/26**	**02/13/26**	**0%**
Audio Branch (Wav2Vec 2.0)	02/05	02/09	0%
Visual Branch (MobileNetV3)	02/10	02/13	0%
Implementation Phase 2	**02/16/26**	**03/13/26**	**0%**
Fusion and Training	**02/16/26**	**02/27/26**	**0%**
DiMoDif Fusion Integration	02/16	02/20	0%
Model Training and Tuning	02/23	02/27	0%
Demo Software	**03/02/26**	**03/13/26**	**0%**
Backend API Development	03/02	03/06	0%
Web Interface & Input Handling	03/09	03/13	0%
Analysis and Evaluation	**03/16/26**	**03/27/26**	**0%**
System Evaluation	**03/16/26**	**03/20/26**	**0%**
Metric Analysis	03/16	03/18	0%
Error Analysis	03/19	03/20	0%
Results Documentation	03/23	03/27	0%
Finalisation	**03/30/26**	**05/08/26**	**0%**
Demo Stabilisation	03/30	04/10	0%
Final Dissertation Write-up	04/13	04/30	0%
Showcase (Optional)	04/20	04/22	0%
Buffer time	04/23	05/07	0%
Final Submission	05/08	05/08	0%



References

- Baevski, A., Zhou, H., Mohamed, A. and Auli, M. (2020) 'wav2vec 2.0: A framework for self-supervised learning of speech representations'. Available at: https://arxiv.org/abs/2006.11477 .
- Cai, Z., Kuckreja, K., Chuchra, A., Khan, M. H., Gedeon, T. and Dhall, A. (2025) 'AV-Deepfake1M++: A large-scale audio-visual deepfake benchmark with real-world perturbations', Proceedings of 33rd ACM International Conference on Multimedia (ACMMM '25). Available at: https://arxiv.org/pdf/2507.20579 .
- Chesney, B. and Citron, D. (2019) 'Deep fakes: A looming challenge for privacy, democracy, and national security', California Law Review, 107, pp. 1753–1820. Available at: https://www.californialawreview.org/print/deep-fakes-a-looming-challenge-for-privacy-democracy-and-national-security .
- Dolhansky, B. et al. (2020) 'The deepfake detection challenge (DFDC) dataset'. Available at: https://arxiv.org/pdf/2006.07397 .
- Howard, A. et al. (2019) 'Searching for MobileNetV3'. Available at: https://arxiv.org/abs/1905.02244 .
- Rossler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J. and Niessner, M. (2019) 'FaceForensics++: Learning to detect manipulated facial images', Proceedings of the IEEE International Conference on Computer Vision, pp. 1–11. Available at: https://arxiv.org/pdf/1901.08971 .
- Yi, J., Wang, C., Tao, J., Zhang, X., Zhang, C. Y. and Zhao, Y. (2023) 'Audio deepfake detection: A survey'. Available at: https://arxiv.org/abs/2308.14970 .